KULLIYYAH OF ENGINEERING (KOE)

MECHATRONICS SYSTEM INTEGRATION (MCTA3203)

SEMESTER 1, 25/26

SECTION 1

PROJECT REPORT WEEK 5

TITLE: Controlling a DC Motor Using L298P Motor Driver Shield and GPIO

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ABSTRACT

This experiment demonstrates the use of an L298P Motor Driver Shield to control the direction and speed of a DC motor through Arduino GPIO pins. By implementing Pulse Width Modulation (PWM), the speed of the motor is varied according to duty cycle percentage. The experiment also involves observing the motor's response to changes in PWM values and direction control logic. The results verify that PWM effectively regulates motor speed and that the L298P allows bidirectional motion using simple logic signals.

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1.0 INTRODUCTION

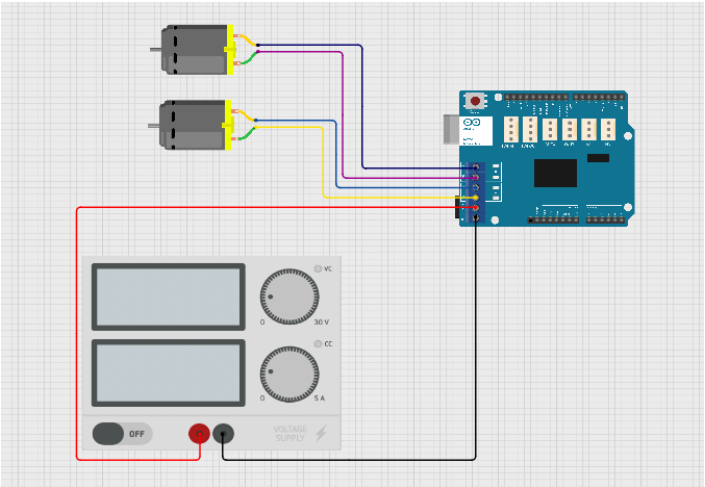
DC motors are widely used in automation systems due to their simplicity and controllability. The L298P Motor Driver Shield acts as an interface between low-power control circuits and higher-current DC motors. It uses an H-bridge configuration that allows bidirectional control by reversing current flow through the motor. Pulse Width Modulation (PWM) provides an efficient way to adjust the motor's speed by controlling the average voltage supplied. This experiment aims to understand the L298P shield's operation and apply PWM through Arduino to achieve smooth speed and direction control.

2.0 MATERIALS AND EQUIPMENT

- Arduino UNO
- L298P Motor Driver Shield
- DC Motor (6V–12V)
- External Power Supply (9V or 12V)
- Jumper Wires
- Breadboard

3.0 EXPERIMENTAL SETUP

- Dc motor A —--> Motor A slot on Arduino motor shield
- Dc motor B —--> Motor B slot on Arduino motor shield
- Power supply cathode —--> VCC on Arduino motor shield
- Power supply anode—--> GND on Arduino motor shield



4.0 METHODOLOGY

1. Connect the L298P Shield to the Arduino UNO and attach the DC motor to the Motor A terminal.
2. Supply 12V to the shield and ensure the ground connection is common with Arduino.
3. Upload the following Arduino code using the Arduino IDE:

```
// Motor A pin definitions
const int ENA = 5; // PWM pin for speed control
const int IN1 = 4; // Direction
const int IN2 = 4; // Direction

void setup() {
pinMode(ENA, OUTPUT);
pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);
}

void loop() {
// Move forward
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
analogWrite(ENA, 255); // Speed: 0-255
delay(2000);
}
```

4. Observe the motor's direction and speed.
5. Modify PWM values (e.g., 64, 128, 255) and record the observed speed changes.
6. Record a slow-motion video and count the number of rotations within a 2-second interval to calculate the approximate RPM.

5.0 DATA COLLECTION

The speed of the DC motor was determined using a slow-motion video recording. The video was played back frame by frame to count the number of full rotations completed within a 2-second interval. The rotations were then converted to revolutions per minute (RPM) using:

$$RPM = \frac{2\ rotations\ in\ 2s \times 60}{2}$$

The results are summarized below:

PWM Value	Direction	Rotations in 2 s	Calculated RPM	Observation
255	Forward	12	(12 × 60)/2 = **360 RPM**	Very fast, smooth rotation
128	Forward	7	(7 × 60)/2 = **210 RPM**	Moderate speed
255	Reverse	11	(11 × 60)/2 = **330 RPM**	Fast rotation, opposite direction
64	Reverse	4	(4 × 60)/2 = **120 RPM**	Slow rotation, easily visible

6.0 DATA ANALYSIS

The collected data shows a clear trend between the PWM duty cycle and the motor's rotational speed. The motor's speed was estimated based on the number of full rotations within a 2-second window recorded using a slow-motion video. This approach was necessary because a tachometer was not available in the laboratory.

From the results, the rotational speed (in RPM) increased significantly when the PWM value was raised. For instance, at a full PWM signal of 255, the motor achieved approximately 360 RPM, while at half the PWM value (128), the speed dropped to around 210 RPM. When reduced further to 64, the motor speed decreased to 120 RPM. This nearly linear relationship between PWM value and RPM reflects the direct influence of the duty cycle on the average voltage supplied to the motor.

Additionally, the motor showed slightly lower speed in reverse rotation (330 RPM) compared to forward rotation (360 RPM) at the same PWM level. This minor discrepancy can be

attributed to mechanical resistance and unequal current flow through the motor's brushes or connections. Such variations are common and do not significantly affect overall performance.

This data confirms that PWM is an effective and efficient method to control motor speed digitally, providing smooth and precise adjustment without excessive heat dissipation that would occur in analog voltage control.

7.0 RESULTS

The experiment successfully verified both speed and direction control using the L298P Motor Driver Shield.
The following key observations were made:

1. Speed Control:
   Increasing the PWM duty cycle increased the average voltage applied to the motor, thereby increasing its speed. The results demonstrated a proportional relationship between PWM value and RPM.
2. Direction Control:
   The direction of the motor changed immediately when the input logic (IN1 and IN2) was reversed. This shows that the L298P's H-bridge configuration correctly handled current reversal through the motor terminals.
3. Braking Behavior:
   When both IN1 and IN2 were set HIGH, the motor stopped abruptly due to both terminals having the same potential — known as electrical braking. This behavior matches the expected theory of H-bridge operation.
4. Measurement Technique:
   Despite the absence of a tachometer, the slow-motion video method proved sufficient for estimating relative speed differences. Although not perfectly precise, it effectively demonstrated how PWM affects the motor's behavior.

In summary, the results validated the theoretical understanding of PWM motor control, showing that higher duty cycles produced higher speeds, and direction control was fully functional.

8.0 DISCUSSION

This experiment demonstrates several core mechatronics principles, namely **electronic motor control**, **PWM modulation**, and **H-bridge direction switching**.

PWM Functionality and Efficiency

PWM allows a digital system like the Arduino to simulate analog voltage control efficiently. Instead of dissipating excess energy as heat, it rapidly switches between ON and OFF states, creating an average voltage proportional to the duty cycle.
This ensures:

- High efficiency with minimal power loss.
- Smooth speed control even at lower voltages.
- Compatibility with microcontrollers that output digital signals.

The results clearly show this effect — when PWM was reduced from 255 to 64, the RPM dropped proportionally, demonstrating fine-grained control of motor speed using digital logic.

Direction Control and Braking

The experiment also confirmed how H-bridge circuits enable bidirectional rotation. Reversing the logic inputs (IN1 and IN2) reversed current flow, changing rotation direction.
Setting both IN1 and IN2 HIGH resulted in a strong braking effect. This is due to both motor terminals being connected to the same voltage level, preventing current flow and dissipating kinetic energy as heat in the windings, thus stopping the motor rapidly.

Experimental Accuracy and Limitations

The main limitation of this experiment was the absence of precise measuring equipment. The **slow-motion video method**, though practical, introduced human error during rotation counting and timing. However, it still provided a reasonable comparison between PWM settings.

Other possible inaccuracies include:

- Motor load variations or internal friction.
- Slight voltage drops due to wire resistance or loose connections.
- Power supply fluctuations affect motor performance.

Despite these factors, the experimental trend aligns perfectly with theoretical expectations, confirming that the system operated correctly.

Applications

The principles learned here are widely used in robotics, electric vehicles, and automation systems, where PWM motor control provides efficient motion regulation. The L298P shield can also be integrated into dual-motor control systems for differential drive robots, conveyor mechanisms, and precision control applications.

Questions: Explain the function of the ENA and ENB pins.

Answer: ENA & ENA (Enable motor A/B) pins are used to enable or disable the corresponding motor driver channel (A or B). When the pin is set to HIGH, the corresponding channel is enabled, allowing current to flow to the motor. The ENA and ENB is also where the PWM signal is applied to control the speed of the motor.

Questions: Describe the reason PWM is used for speed control.

Answer: PWM or Pulse Width Modulation is a technique for achieving analog control results using a digital signal. A PWM signal is defined by 2 main characteristics that are crucial for speed control which is the duty cycle and frequency. There is a few reason why PWM is used to control motor speed instead of analog control such as its very low power loss since the switch is either fully ON (voltage drop is nearly zero) or fully OFF (current is zero) resulting in minimal power converted into heat during switching. PWM control is also proven to be more efficient and have a better low-speed torque since the pulses are full voltage, the motor will maintain higher effective torque even at low average speed.

Questions: Describe the outcome when both IN1 and IN2 are set HIGH.

Answer: IN 1 and IN 2 are both the direction control pins for motor A that control the motor to move in different directions (forward and reverse). When both pins are set to HIGH, the motor will receive a signal to go forward and backward at the same time resulting in **braking or short-circuit braking**. This is because both outputs to the motor winding are internally connected to the motor’s power supply, effectively shorting the motor's terminal together across the supply voltage when both IN 1 and IN 2 are HIGH.

Questions: Explain how braking can be implemented using the L298P.

Answer: As explained above, the most common and effective way to implement braking using L298P is by setting both input pins, IN 1 and IN 2 for motor A, IN 3 and IN 4 for motor B to HIGH simultaneously resulting in dynamic braking. Setting both input pins to LOW can also stop the motor (Coasting) but slower than the HIGH method.

Questions: Modify the code to control two DC motors simultaneously.

Answer:

```
// Motor A pin
const int ENA = 5; // PWM pin for speed control
const int IN1 = 4; // Direction
const int IN2 = 4; // Direction

//Motor B pin
const int ENB = 6; // PWM pin for speed control
const int IN3 = 7; // Direction
const int IN4 = 7; // Direction


void setup() {
pinMode(ENA, OUTPUT);
pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);
pinMode(ENB, OUTPUT);
pinMode(IN3, OUTPUT);
pinMode(IN4, OUTPUT);
}
void loop() {
// Move forward
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
analogWrite(ENA, 64); // Speed: 0-255

digitalWrite(IN3, HIGH);
digitalWrite(IN4, LOW);
analogWrite(ENB, 64); // Speed: 0-255
delay(5000);
// Move backward
digitalWrite(IN1, LOW);
digitalWrite(IN2, HIGH);
analogWrite(ENA, 64);

digitalWrite(IN3, LOW);
digitalWrite(IN4, HIGH);
analogWrite(ENB, 64);
delay(2000);
// Stop
analogWrite(ENA, 0);
delay(1000);}
```

9.0 CONCLUSION

The experiment has successfully demonstrated the working principle and functionality of the L298P Motor Driver Shield for controlling a DC motor's speed and direction using an Arduino.
The implementation of PWM proved to be an efficient and precise method of controlling speed, while direction control through digital logic inputs (IN1 and IN2) functioned exactly as expected.

Key findings include:

- Motor speed is directly proportional to the PWM duty cycle.
- Reversing IN1 and IN2 changes the direction of rotation.
- Setting both IN1 and IN2 HIGH creates an electrical braking effect.
- Even without a tachometer, the slow-motion video method provided acceptable accuracy for speed estimation.

The experiment fully achieved its objectives and provided valuable insights into digital motor control systems — a fundamental skill in mechatronics and embedded design.

The hypothesis that PWM can precisely control motor speed while maintaining efficiency and direction control was proven correct.

10.0 RECOMMENDATIONS

To improve future experiments and obtain more accurate and meaningful results, the following recommendations are proposed:

1. **Use a Tachometer or Encoder:**
   A digital tachometer or rotary encoder should be used to measure RPM accurately and eliminate human counting error.
2. **Implement Serial Data Logging:**
   Adding serial output to the Arduino program can display and record PWM values, directions, and calculated speeds in real-time for better data analysis.
3. **Calibrate Power Supply:**
   Ensure that the external power source provides a stable voltage, as fluctuations can affect motor speed and consistency.

4. **Expand to Dual-Motor System:**
   Modify the code to control two motors simultaneously (Motor A and Motor B) to simulate robot movement or differential drive systems.
5. **Improve Braking Control:**
   Experiment with both **coasting stop** (ENA = LOW) and **short braking** (IN1 = IN2 = HIGH) to analyze stopping performance under different conditions.
6. **Enhance Measurement Method:**
   If slow-motion video is used again, add a visible shaft marker or colored tape to improve rotation counting accuracy.
7. **Add Feedback Control:**
   Implement a **closed-loop control system** (using a sensor or encoder) to automatically regulate speed according to set values, introducing real-world control concepts.

By applying these improvements, future students can achieve higher precision and deepen their understanding of motor control systems.

11.0 REFERENCES

References
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L298P_MotorDriverShield
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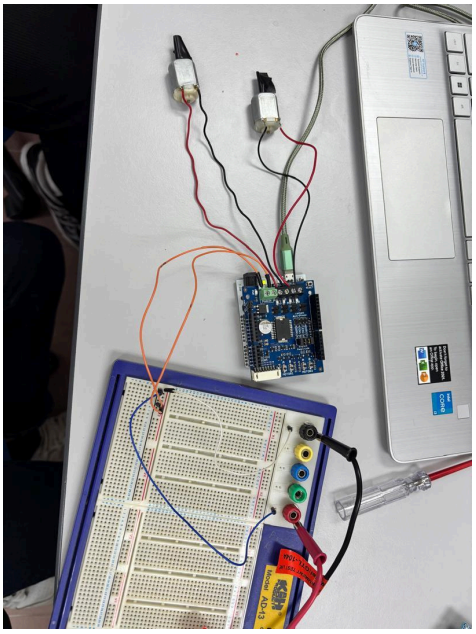
How to Use L298P drive shield: Examples, Pinouts, and Specs

12.0 APPENDICES



13.0 ACKNOWLEDGEMENT

We would like to express our gratitude to my lab instructor, Dr Zulkifli bin Zainal Abidin, for his guidance during the experiment. Special thanks to my groupmates and peers for their collaboration and help during circuit setup and troubleshooting.

14.0 STUDENTS DECLARATION

CERTIFICATE OF ORIGINALITY AND AUTHENTICITY

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specific in references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or a person.

We hereby certify that this report has not been done by only one individual and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We, therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

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Contribution: : Results, discussion, conclusion